XUEWU CAI

Trading Systems, Operational Risk & Technology Controls | FIX, Java, Python, AWS | Regulatory Reporting

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PROFESSIONAL SUMMARY

Technology and risk leader with more than 20 years designing, operating, and overseeing the systems that support electronic trading, market data, and financial risk reporting. Built and led development of a multithreaded, fault-tolerant trading platform handling multiple asset classes and more than $3 billion in capital, and later produced SEC, CFTC, and OPERA regulatory reporting as a risk engineer. Today, as Director, Financial Engineering at Freddie Mac, directs large-scale analytics on AWS and IBM Symphony for credit-risk-transfer programs covering more than $1.6 trillion in mortgage loans, with responsibility for model governance, controls, and audit response. Combines hands-on engineering credibility with the examination mindset needed to assess reliability, capacity, security, and recovery practices at regulated entities.

CORE COMPETENCIES

Trading & Market Systems: electronic trading platforms, FIX connectivity, order routing, market data feeds, time-series infrastructure
System Safeguards: reliability and failover design, capacity and performance planning, business continuity and recovery, change and release controls
Risk & Controls: operational and model risk, control testing, reconciliation, audit and regulatory examination response
Regulatory Reporting: SEC, CFTC, and OPERA reporting; SOFR transition; CECL; capital and stress testing
Engineering: Java, Python, C/C++, MATLAB, SQL, Docker, AWS, IBM Symphony, QuantLib, automated testing
Stakeholder Communication: examination findings, technical documentation, executive briefings, cross-agency coordination

PROFESSIONAL EXPERIENCE

Director, Financial Engineering — Freddie Mac
May 2014 – Present

McLean, VA | Credit Risk Transfer (STACR / ACIS) | Full-time

- Direct large-scale financial analytics on AWS and IBM Symphony for STACR and ACIS programs covering more than $1.6 trillion in mortgage loans, with production controls, reconciliation, and audit-ready documentation.
- Own model and analytics governance for transaction-critical systems: version control, validation, change management, and response to internal audit and regulatory inquiries.
- Led the LIBOR-to-SOFR transition across CRT models and analytics — translating regulatory requirements into engineering controls, test plans, and remediation.
- Deliver transaction economics and risk measurement (duration, convexity, OAS, KRDs, capital, ROE) that depend on reliable, high-volume automated cash-flow and valuation systems.
- Apply generative AI and agentic engineering to accelerate delivery while maintaining control, traceability, and review standards.
- Partner with risk, finance, legal, and technology stakeholders to prioritize remediation and align system changes with policy requirements.

Research Engineer / Quantitative Finance Engineer / Senior Software Engineer
Mar 2003 – Apr 2014

Campbell & Company | Washington, DC–Baltimore, MD | Full-time

Progressive engineering roles across trading systems, market data, risk reporting, and derivatives.

- Led architecture and development of a multithreaded, fault-tolerant electronic trading platform (Java, Portware, FIX) supporting concurrent strategies, orders, and multiple asset classes, with more than $3 billion in capital and tens of millions of dollars in slippage savings.
- Built a Java-based intraday cash equity trading system on Caché and time-series data infrastructure and market data feeds in MFC/C++, directly comparable to evaluating automated trading and data-reporting environments.
- Designed and implemented risk reporting — factor analysis, scenarios, stress tests, duration, and margin — and produced SEC, CFTC, and OPERA regulatory reporting.

Senior Software Engineer — Aquilent
Jun 1999 – Feb 2003

Full-time | Java/Linux integration and data systems

- Built Java/Linux sensor-integration systems, including embedded Linux deployments, JNI hardware interfaces, serial communications, XML instrument definitions, and dynamic sensor discovery.

EDUCATION & PROFESSIONAL DEVELOPMENT

M.S., Computer Science — University of Maryland, Baltimore County
M.S., Computer Science & Engineering — Shanghai Jiao Tong University
B.S., Applied Mathematics — Dalian University of Technology
Professional development: CAIA (Chartered Alternative Investment Analyst) Level I

LANGUAGES

Chinese (native or bilingual) | English (full professional)